

Eva Abou Harb

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Instructor: Adelaja Odujinrin

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Selection of Labs

For this final exam reflection, I selected three labs that cover different areas of cybersecurity and are different from the labs I selected for my midterm. The first lab is **Phishing Analysis Fundamentals** from TryHackMe, which focuses on identifying phishing emails, analyzing email headers, and understanding social engineering attacks. The second lab is **Identification and Scoping**, also from TryHackMe, which focuses on the early stages of incident response, including identifying security incidents, collecting evidence, and determining the scope of a compromise. The third lab is **Build a Secure Agent with Model Armor and Identity** from Google Codelabs, which focuses on securing AI agents using access control, prompt injection protection, and cloud security best practices.

These labs cover three distinct areas: phishing analysis, incident response, and AI security. Together, they helped me better understand both traditional cybersecurity threats and modern security challenges involving artificial intelligence systems.

Knowledge Gained

From the **Phishing Analysis Fundamentals** lab on TryHackMe, I learned how phishing emails are structured and how attackers use email to trick users into revealing sensitive information or clicking malicious links. The lab introduced the anatomy of an email address, including the username, domain, and the purpose of the @ symbol. I also learned how email delivery works using protocols such as SMTP, POP3, IMAP, and DNS. Understanding how

emails travel from sender to recipient helped me better understand how attackers abuse these systems.

Another important concept covered in the lab was email header analysis. I learned how to inspect raw email files and use message source data to identify useful information such as the sender address, subject line, reply-to address, X-Originating-IP, and suspicious attachments. I also learned how to analyze HTML email bodies and decode base64 attachments to safely reconstruct files for investigation. The lab also explained different types of phishing, including spam, phishing, spear phishing, whaling, smishing, vishing, and Business Email Compromise (BEC). Through the practical tasks, I identified spoofed organizations, suspicious domains, originating IP addresses, and malicious email indicators.

In the **Identification and Scoping** lab, I learned how the identification phase forms the foundation of the incident response process. This phase requires a combination of people, processes, and technology to recognize security alerts, understand their severity, and report incidents properly. I learned that identifying a security incident quickly helps reduce damage and improves response time. The lab also explained how employees across an organization, not only the IT team, must report suspicious behavior and follow the correct procedures.

The lab also introduced the importance of scoping security incidents. I learned how organizations use tools such as the Asset Inventory and the Spreadsheet of Doom (SoD) to identify affected systems and track Indicators of Compromise (IoCs) such as malicious domains, IP addresses, email addresses, and file hashes. These tools help investigators understand the extent of a compromise and organize evidence efficiently. I also learned about the feedback loop between identification and scoping, where collected evidence leads to new pivot points for

investigation. This process improves the overall effectiveness of incident response and helps security teams make better decisions.

In the **Build a Secure Agent with Model Armor and Identity** lab from Google Codelabs, I learned how AI systems can introduce serious security risks if they are not properly protected. The lab focused on threats such as prompt injection attacks, sensitive data disclosure, unauthorized access, and rogue agent actions. I learned that attackers can manipulate AI agents by giving malicious prompts that attempt to override instructions or expose restricted information.

The lab introduced Google Cloud Model Armor, which filters prompt injections, sensitive data such as Social Security numbers and credit card numbers, harmful content, and malicious URLs. I created a Model Armor template and a Model Armor guard that checked both user input and model output before processing. Another important concept was Agent Identity, which uses IAM permissions to restrict what the AI agent can access. Instead of trusting the model to behave correctly, the agent was only given access to the `customer_service` BigQuery dataset and was blocked from accessing the admin dataset. I also performed red team testing to verify that prompt injection attempts were blocked, sensitive data was filtered, and unauthorized access requests were denied.

Application in a Professional Setting

The knowledge gained from these labs can be applied in several cybersecurity roles. The skills learned from the Phishing Analysis Fundamentals lab are especially useful for SOC analysts, security analysts, and incident responders. In a real workplace, phishing remains one of the most common attack methods used by cybercriminals. Security professionals must be able to analyze suspicious emails, inspect email headers, identify spoofed domains, and prevent users

from clicking malicious links or opening harmful attachments. These skills help organizations prevent credential theft, malware infections, and unauthorized access.

The knowledge gained from the Identification and Scoping lab is essential for incident response teams and security operations centers. When a security alert is triggered, professionals must investigate the issue quickly, gather logs, document findings, identify affected systems, and determine the full scope of the incident. Using tools such as asset inventories and IoC tracking spreadsheets helps security teams work efficiently and communicate clearly during an active incident. This process is critical for containment, recovery, and preventing similar incidents in the future.

Finally, the knowledge gained from the Build a Secure Agent with Model Armor and Identity lab is increasingly important as organizations adopt AI-powered systems. Security professionals must understand how AI agents can be manipulated through prompt injection, data leakage, and unauthorized access attempts. By implementing strong access controls, input and output filtering, logging, and red team testing, organizations can reduce the risks associated with AI systems. This is especially important for companies using AI assistants connected to internal databases or customer information.

These labs showed me that cybersecurity is constantly evolving, and professionals must be prepared to secure both traditional systems and modern AI applications. Hands-on practice in these labs helped me better understand how theory applies to real-world security challenges.

Lab References

Build a Secure Agent with Model Armor and Identity - Google Codelabs

<https://codelabs.developers.google.com/secure-customer-service-agent/instructions>

Identification and Scoping - TryHackMe

[TryHackMe | Identification & Scoping](#)

Phishing Analysis Fundamentals – TryHackMe

[TryHackMe | Phishing Analysis Fundamentals](#)